

Aussie Primary Academy

Year 6 Science – Chemical Sciences: Reversible & Irreversible Changes

I can: classify changes to materials as reversible or irreversible and explain why!

Name: _____

Date: _____

Page 1 – Let's Explore!

ACARA V9.0 · AC9S6U02

Big Idea

When materials change, some changes can be **undone** (reversible), like melting chocolate that can be cooled back into a solid. Other changes create a **new substance** that cannot be undone (irreversible), like toast that cannot turn back into bread.

1 Vocabulary Bank

reversible

irreversible

new substance

chemical change

physical change

rust

dissolve

burn

2 Sort into reversible or irreversible

melting ice, burning wood, folding paper, baking a cake, dissolving sugar, rusting iron, freezing water, cooking an egg

Reversible

Irreversible

3 Explain why each is irreversible

a) Burning wood — _____

b) Cooking an egg — _____

4 True or False

a) Melting and freezing are reversible changes. True / False

b) Rusting creates a brand new substance. True / False

c) All changes involving heat are irreversible. True / False

- 5 Draw and label an example of a reversible AND an irreversible change you have seen at home

Reversible



Irreversible



- 6 Investigate: Test a material change

Choose a change to test at home (e.g. dissolving salt in water). Can it be reversed? How?



- 7 Challenge Question

Is dissolving sugar in water reversible or irreversible? Explain how you know.

Great Science Thinking!

High-Five Tips for Parents & Teachers

- Cook or bake together and pause to ask 'can we undo this step?' at each stage — a genuinely hands-on way to explore the concept.
- Rusting old outdoor metal objects (a fence, tools) are a great slow-motion example of irreversible change to point out on a walk.
- Avoid the common mix-up: melting/freezing is reversible, but burning/cooking creates a new substance and cannot be undone. Ask 'what evidence tells you a new substance was made?' (colour change, smell, gas produced, can't get the original back).
- Praise reasoning, not just the right answer — 'why do you think that?' builds deeper scientific understanding than a yes/no check.

✓ Answer Key

2. Reversible: melting ice, folding paper, dissolving sugar, freezing water. Irreversible: burning wood, baking a cake, rusting iron, cooking an egg
4. a) True b) True c) False
7. Reversible — the sugar can be recovered by evaporating the water, no new substance is formed